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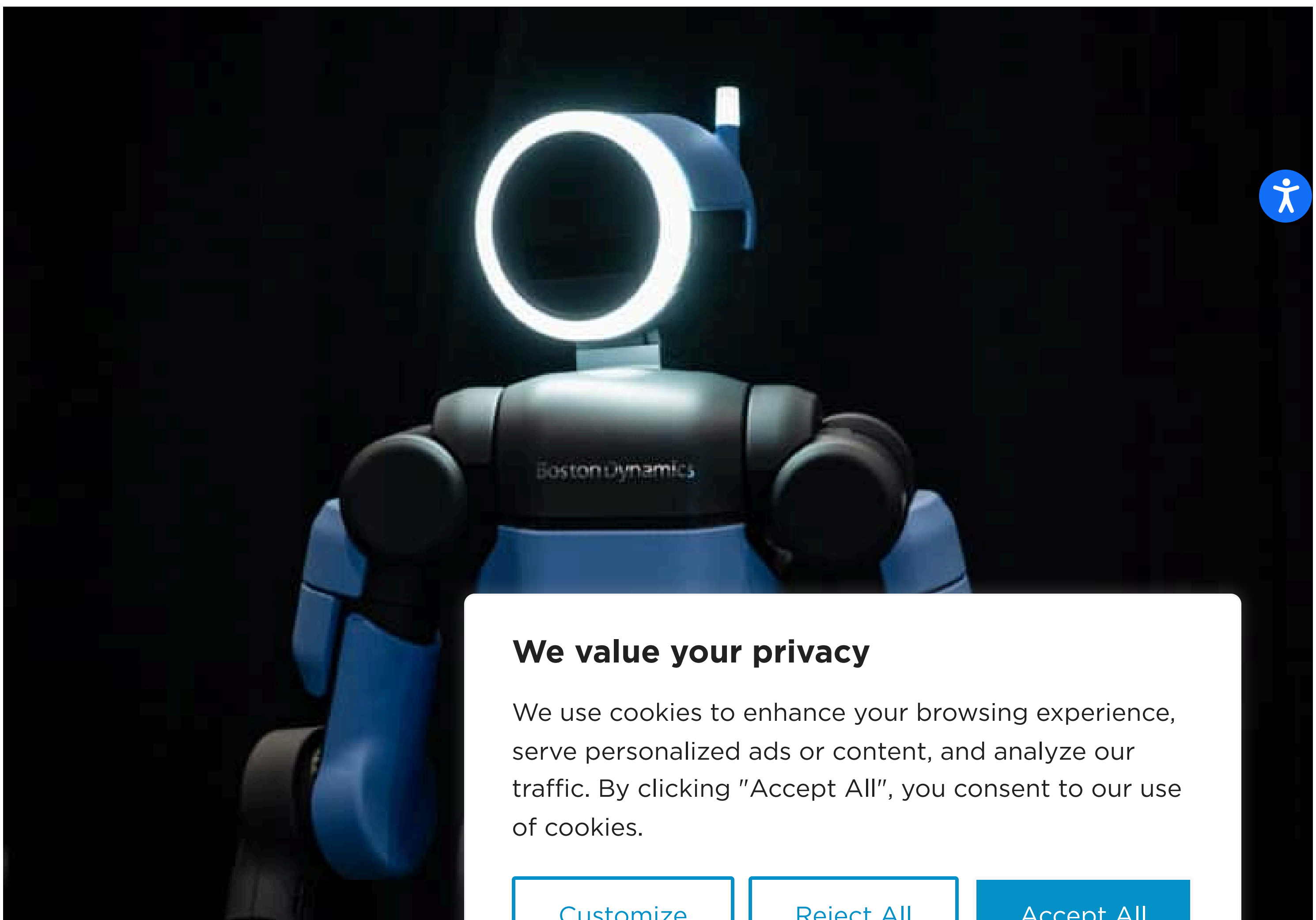
Atlas' Evolution From Research Robot to Industrial Humanoid



workplaces. Most importantly, Atlas is intelligently designed for scale: learned behaviors can be redeployed across fleets, allowing robots to continuously improve and deploy new tasks in less than a day.

We didn't get to this point overnight. Boston Dynamics lead the industry in humanoid research and development, showcasing Atlas' agility, balance, and strength across parkour courses and dance performances. We built the tools and the expertise in full-body control, perception, and complex manipulation. Our commercial journey with Atlas didn't start in 2026 with the announcement of the product version; it started years earlier with the ambition to push the art of the possible and the hands-on expertise to deliver real value.

Looking back at our milestones so far shows how Boston Dynamics reshaped Atlas into the world's leading enterprise humanoid.

From Prototype to Production in 2026

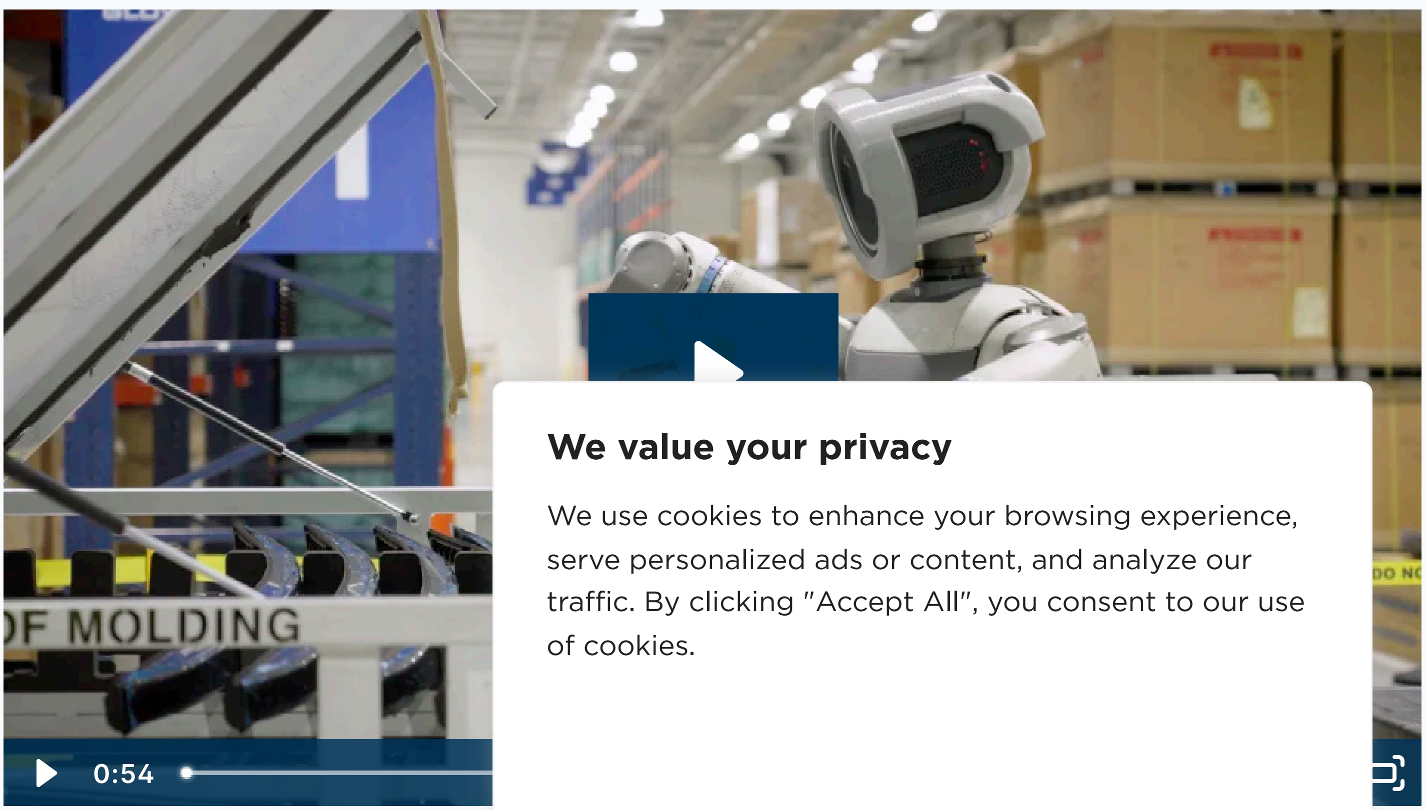
In January 2026, Boston Dynar
ready Atlas—the result of deca
revolutionize industrial automa
robot is strong, agile, and adap

It's easy to use, easy to integra ... ain
for new tasks. It's designed to b ... an

We are already manufacturing the product version of Atlas today, with deployments scheduled this year at Hyundai and Google DeepMind. At the same time we are continuing humanoid robot development and expanding Atlas' generalist capabilities, training the fleet with RL and foundation models.

Learning, Perception, and Autonomy in the Factory and the Lab in 2025

In 2025 our team was laser-focused on our first application: part sequencing in automotive manufacturing. Part sequencing allowed us to tackle core problems working towards a general purpose humanoid, including task diversity, behavior complexity, and environmental complexity. But it also represented a challenge with real ROI behind the solution.



Our team married our AI and controls expertise with practiced hardware and reliability development. We focused on hardening Atlas' performance in key areas:

- Perfecting Atlas' perception with 2D and 3D awareness of its surroundings and the objects it's interacting with to support reliable autonomous operations and AI-driven generalizable behavior in unpredictable environments
- Iterating on gripper design to get the right mix of strength, fine motor skill, and durability
- Practicing a range of grasps on our manipulation test stand, using RL policies trained using NVIDIA workflows
- Applying large behavior models for full-body control and real-time adaptability
- Creating fluid, natural gaits and movement for Atlas, using motion capture and animation RL pipelines developed with RAI Institute

Application readiness served as the primary measure of progress. In the fall, we put Atlas to the test at the Hyundai metaplant in Georgia, inviting 60 Minutes to join us for Atlas' first deployment.

Starting the Commercial Journey with an Electric Humanoid in 2024

The turning point in Atlas' evol

electric Atlas. This marked a de

commercial journey.

Electric actuation was importa

improved energy efficiency, bu

We started proving that Atlas could work autonomously, bringing together advanced mobility and advanced intelligence. We demonstrated the robot executing simple handling tasks with irregular parts, using a machine learning (ML) vision model to detect and localize the environment, specialized grasping policy, and continuous awareness of manipulated objects.

By the end of 2024, Atlas was positioned to move from proof of concept into industrial use. But, true to our roots, it could still do a backflip.

The Research
Atlas Possible

This research legacy is more than flashy demos. Our humanoid robot development work with hydraulic Atlas produced insights into control, balance, and whole-body motion, directly informing later designs and product development. This foundation informed the evolution of Atlas—taking the hardest challenges head on whether in the lab or on the factory floor.

What The Atlas Robot Evolution Reveals

Our journey with Atlas mirrors our journey with the Spot® robot and the evolution of our Stretch® robots—from prototypes to pilots to scaled deployments and ROI. Looking back, Atlas' journey tells a clear story:

- Successful productization and real-world impact are about customer value, with a focus on problems worth solving rather than those that simply look impressive.
- Robotics and AI development require patience, persistence, and collaboration, with breakthroughs compounding over time.
- Intelligent software and powerful hardware go hand in hand. The usefulness of a humanoid robot is determined by intelligence and autonomy, but also robustness, serviceability, and scalabilit
- The humanoid form factor s
  performance or mobility

To learn more about how we m
webinar on hardware and indu

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As Atlas begins its life beyond the lab, its evolution continues — informed by real environments, real constraints, and real outcomes.The next chapters won't be written by Boston Dynamics alone, but will grow in partnership with the customers deploying Atlas in factories, warehouses, and workplaces where adaptability matters every day.

Start a conversation with us to learn how Atlas is being applied in real-world environments.

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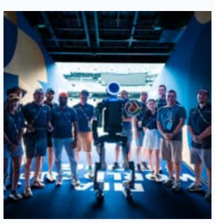

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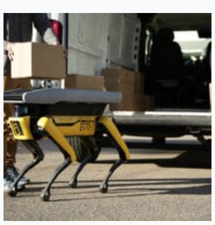

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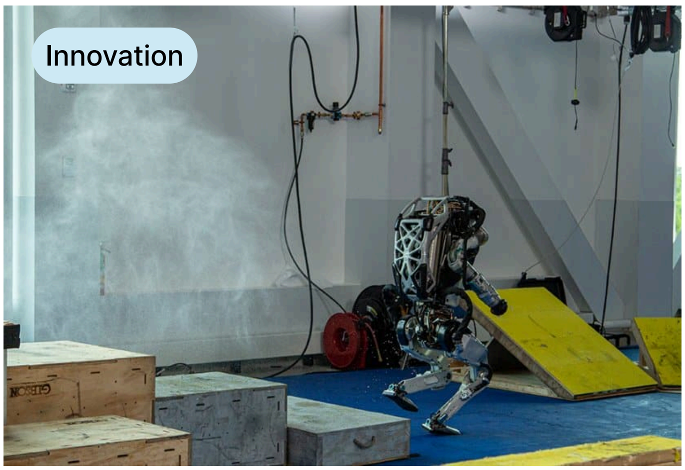

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